

Smart Urban Mobility & Traffic Intelligence Platform

An End-to-End Tourism & Culture Analytics Platform for India

TEAM - 1

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Project Overview & Objectives

Context

India's tourism data remains fragmented across ASI, state tourism boards, and the Ministry of Tourism. Urban Pulse unified this landscape across four milestones from clean data foundations through map-based analytics, operational intelligence, and equity-aware forecasting.

Core Objectives

- Data Pipeline
Galaxy schema + real government data
- Geospatial & Weather Intelligence
Demand pattern visualization
- Mobility & Traffic Intelligence
Visit, booking, last-mile transport intelligence, congestion patterns and modal shifts
- Equity & Forecasting
Transit equity measurement, demand forecasting, and anomaly detection
- Unified Executive Platform
One deployable, user-ready dashboard

Key Metrics

999

Tourist Places Tracked

15.2M

Foreign Tourist Arrivals

279

Festival Records

24,831

Trips in Active Dashboard Window

Tourism Revenue

Rs 3,26,867 Cr

Heritage Zones

6 Total Analysis Zones

Mobility Access Index

58.12 Average Score

Simulated Modal Shift

1.21% Modal Shift Index

These metrics provide a consolidated view of India's tourism ecosystem, cultural assets and mobility conditions from raw monument footfall to executive-level platform health

Avg Mobility Access Index of 58.12 signals meaningful room for equity improvement across heritage transit corridors.

Data Sources and Architecture



Data Layers

Unified Architecture

Galaxy Schema on **Supabase**, queried via **Streamlit**, **Power BI**, and **Looker Studio**.

Tourism & Attractions

Ministry of Tourism / IITM reports

Festivals & Funding

data.gov.in DPPH Scheme (2014–2024)

Weather Intelligence

Open-Meteo API - temperature, rainfall, humidity, AQI

Mobility & Transit

Synthetically generated (Milestone 3), anchored to real weather and monument patterns

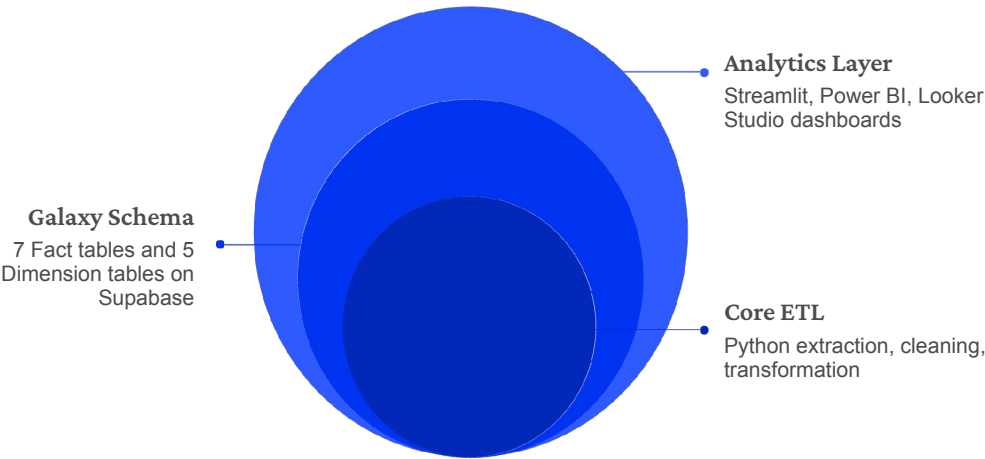
Data Integration and Foundation

ETL Pipeline

Python-driven extraction, cleaning, and transformation into a PostgreSQL cloud database - enabling scalable querying across all platform modules.

Design Rationale

The Galaxy Schema handles multiple concurrent events at varying levels of geographic and temporal detail, making it ideal for India's diverse tourism data landscape.



fact_attractions Site-level visitor data	fact_tourist_bookings Booking and Visit behaviour		fact_festivals Cultural event records	fact_heritage_transport Heritage transit operations	
fact_monthly_tourism Temporal arrival trends	fact_modal_shift_weather Weather driven modal shifts		fact_country_arrivals Foreign origin tracking		
dim_time Chronological filters (year,month,quarter)		Dim_location & location_coordinates Geographical hierarchy (city, state, site)		dim_country International Visitor Origin Data	dim_weather weather condition

Geospatial and Transit Analysis

Transformed raw schema into interactive geospatial intelligence.

Footfall Density Heatmaps

Mapped tourist concentration across India - Delhi, Mumbai, Rajasthan, South India

Origin-Destination Flow Maps

Tracked foreign tourist movement patterns across the country

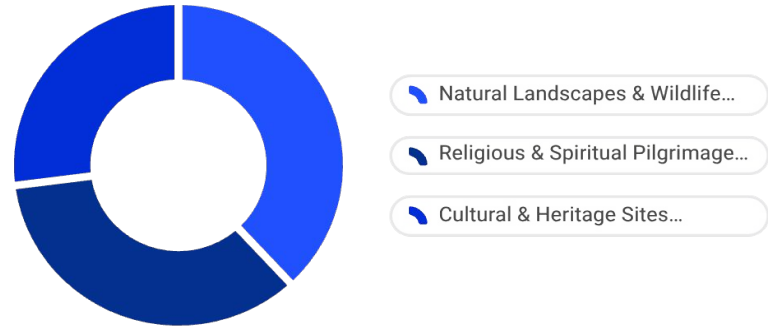
Site Performance Dashboards

Compared capacity vs footfall and festival funding efficiency

3-Level Geographic Drill-Down

National → State → Monument navigation

Attraction Category Breakdown



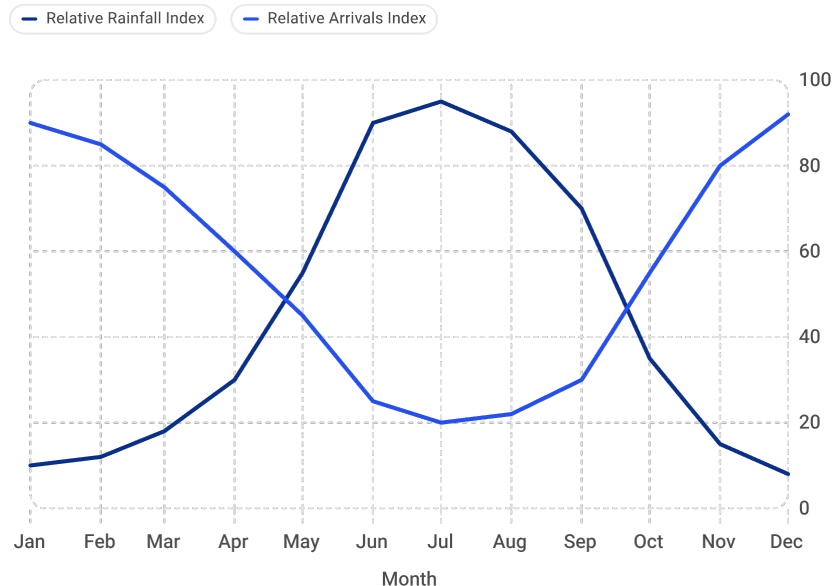
Natural Landscapes & Wildlife lead with **213 sites**, followed closely by Religious & Spiritual Pilgrimages at **195 sites** and Cultural & Heritage Sites at **151 sites** - reflecting India's extraordinary diversity of tourism assets mapped within Urban Pulse.

Weather and Temporal Analysis

Climate Integration

Cross-referenced Open-Meteo weather parameters against historical arrival trends to reveal how India's climate shapes tourism demand throughout the year.

- ❏ **Key Finding:** A distinct inverse relationship exists between monsoon rainfall and foreign tourist arrivals - arrivals drop sharply June - September and recover strongly in winter months.



Dynamic Filtering

users can isolate seasonal patterns dynamically using filters for **Year**, **Quarter**, **Month**, and **Peak/Off-Peak** periods across all dashboards.

1

Peak Season

Oct - Feb: High arrivals, dry weather

2

Shoulder Season

Mar - May: Moderate, warming trend

3

Monsoon Trough

Jun - Sep: Sharp arrival decline

Demand Intelligence and Visit/Booking Analytics

Moved from aggregate footfall to operational, visit-level intelligence via Python/Colab pipeline.

Visit & Booking Intelligence

1000

Total Bookings

575

Completed Visits

24.1

Avg Queue Wait (min)

24.9%

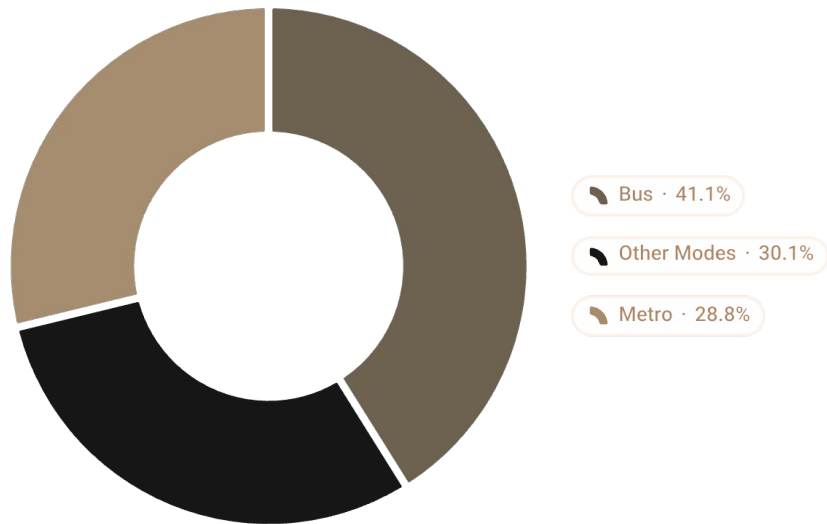
Avg Cancellation Rate

Clear hourly demand peaks identified - enabling targeted staffing and capacity planning recommendations.

Last-Mile Heritage Circuit Analytics

- **195 total trips** across 13 active transport stands
- Supply and demand analysis across heritage circuits

Modal Substitution and Weather Sensitivity



Modal Distribution

Buses dominate last-mile connectivity at **41.1%**, followed by other modes at **30.1%** and Metro at **28.8%** - revealing significant opportunity for metro expansion.

Infrastructure Simulation

A simulated **20% infrastructure improvement** yielded a **1.21% transit shift index**, demonstrating measurable modal rebalancing potential with targeted investment.

Weather Elasticity

Extreme weather events - heavy rain and thunderstorms - reduce shuttle, walk, and e-bike usage while shifting demand toward **cars and metro networks**, requiring dynamic supply adjustments.

Place Based Traffic Intelligence

Destination level heavy congestion comparison across peak and off-peak tourism seasons

Overall Average by Season

31.1%

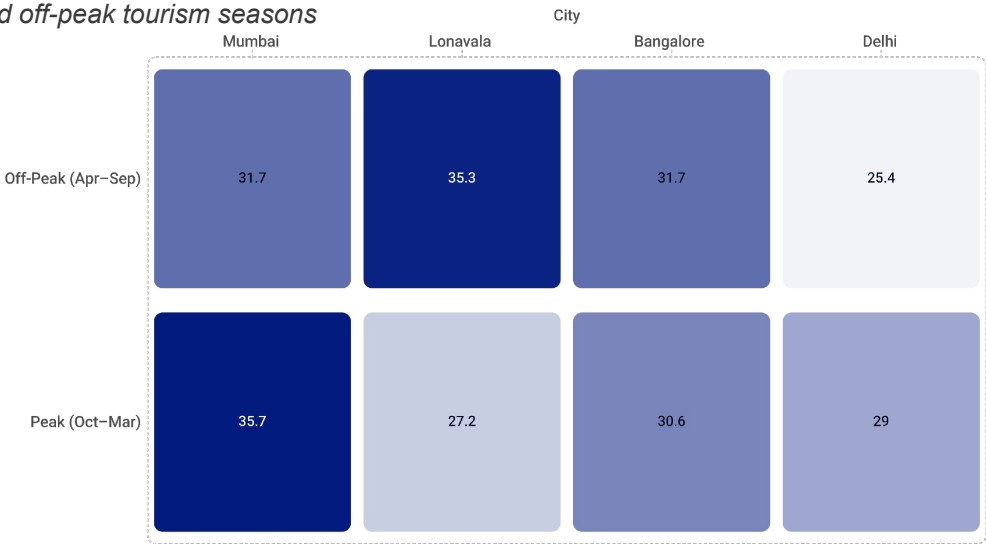
Off-Peak Season
(April - September)

30.5%

Peak Season
(October - March)

35.7%

Mumbai - Peak Season
(Highest recorded city average)



Strategic Insights

Mumbai: Peak-Driven

- 1 Rises from 31.7% to **35.7%** in peak season :the highest single value across all cities and seasons.

Delhi: Consistent Rise

- 3 Grows from 25.4% to **29.0%** peak : moderate but steadily climbing with tourist influx.

Lonavala: Off-Peak Surge

- 2 Inversely peaks at **35.3%** off-season, then drops to 27.2% during peak months : a distinctive counter-seasonal pattern.

Bangalore: Near-Stable

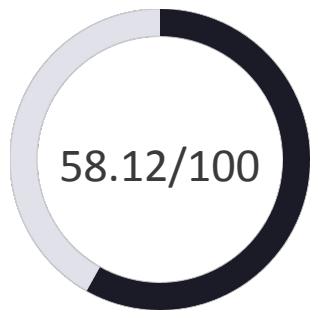
- 4 Slight dip from 31.7% to 30.6% : relatively consistent across both seasons, driven by year-round urban traffic.

□ Data represents place-based and seasonal analysis from historical and simulated records - not real-time monitoring.

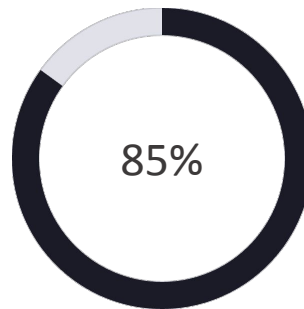
Mobility Access and Equity Dashboard

Measures how equitably tourists can reach heritage sites via public transit.

Monument Accessibility & Equity Index



Avg Mobility Access Index
Across 6 heritage zones analyzed



Transit Performance
Platform-wide fixed reference value

Stands classified as **Well Connected** or **Moderately Connected** based on bus/metro trip density relative to location.

Implementation Insight

Cross-references fact_heritage_transport against dim_location - required no new data collection, only new analytical logic on our existing Galaxy Schema.

Executive Intelligence, Forecasting & Anomaly Detection

24,831

Total Trips (recorded across the platform)

40.0

Avg Travel Time (minutes per trip)

85.0%

On-Time Performance (transit reliability rate)

91.0

Mobility Health Score (out of 100)

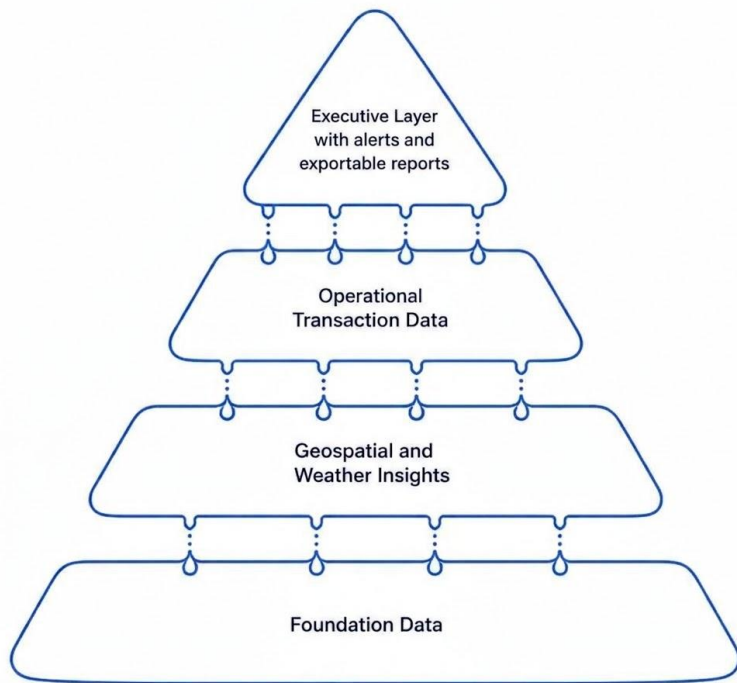
Forecasting & Alerts

- Time-series demand forecasting with confidence bounds
- Automated alerts for unusual changes flagging statistical surge spikes and demand drops

Reporting Suite

- Automated **CSV and PDF export** capabilities for use
- Exportable reports ready for government and tourism board distribution

Integrated Smart City Dashboard



Unified Architecture

- Combines all analytical pillars into a **single executive dashboard**
- Replaces four disconnected module views with one cohesive interface

Cross-Module Synergy

- **Foundation data** feeds all downstream modules
- Geospatial and weather insights inform equity and forecasting logic
- Operational transaction data powers the equity index and demand predictions
- Final executive layers tie everything together into **automated analytical alerts and exportable reports**

Key Results and Benefits

Live Platform Deployment

Fully deployed, GitHub-hosted platform live via public Streamlit URL - accessible anywhere

Comprehensive Cultural Coverage

Analysis of **999 tourist attractions** and **279 festival records** across India

Monsoon Impact Validated

Historical analysis identified lower tourist arrivals during high-rainfall months, while simulated mobility records showed weather related changes in transport choices.

Actionable Equity Index

Uncovers underserved heritage zones for tourism boards, enabling targeted policy intervention.

Automated Intelligence

Automated forecasting and anomaly detection replacing reactive, manual monitoring workflows.

Non-technical Accessibility

Exportable reports making the platform usable by non-technical users at every level.

Limitations, Future Enhancements & Conclusion

Limitations

Milestone 3 booking and transport data is **synthetically generated**, anchored to real weather and monument patterns.

Forecasting uses **historical pattern projection** rather than a trained ML model.

Conclusion

Across four milestones, **Smart Urban Mobility and Traffic Intelligence Dashboard** evolved from a clean data foundation into a genuinely deployable tourism intelligence platform - grounded in real Government of India data, validated by real weather-demand correlations, and structured for real-world adoption by state tourism boards and policymakers.

Future Enhancements

01

Live Traffic API Integration

Replace synthetic data with live transit feed, connect google maps, Mapbox, city traffic sensors or public transport feeds for real tie congestion monitoring.

02

ML-Powered Forecasting

Extend with proper time-series models

03

Unified Single Codebase

Build a true single-codebase unified dashboard for production.

Thank You

Team 1 - Smart Urban Mobility & Traffic Intelligence Dashboard
(TOURISM & CULTURE USE CASE)